Forecasting Sampled Net-Position Peaks: When Do Conditional Quantiles Beat Historical Rules?

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Abstract

Inventory monitoring concerns the largest position reached during a horizon as well as its endpoint. Using an hourly FX client-flow position record and weekly Treasury-bill aggregates, we compare historical rules and conditional quantiles on matched completed-label histories and update schedules. Long-history regression reduces six-step FX pinball loss by 12.7%, while its original Treasury counterpart increases loss by 50.8%. Expanding the Treasury penalty menu reduces that excess to 3.4%, approaching an intercept-only empirical-quantile reference while preserving the FX gain. Across 12 additional archived weekly series, expansion reduces median excess loss from 92.6% to 9.2%, but regression still loses in 34 of 48 cells. Restrictive tuning can therefore explain substantial deterioration without explaining it all. Lower mean score coexists with more breaches, and neither primary comparison dominates across all consistent quantile scores. The analyses are retrospective: FX outcomes were previously inspected, Treasury is replayed by economic date from a downloaded vintage, and the panel shares dealer reporting and overlapping positions.

Keywords: quantile forecasting; dealer positions; sampled path maxima; forecast evaluation.

1 Introduction

Dealer inventories connect customer order imbalances to risk limits and balance-sheet capacity. Inventory affects optimal dealer pricing and observed position adjustment (Ho and Stoll, 1981; Lyons, 1995); dealer capacity also matters for Treasury intermediation (Duffie et al., 2023). A position can become large during a holding interval and then return toward its starting value. Monitoring its largest recorded magnitude therefore asks a different question from predicting the endpoint.

We ask when conditional quantile methods forecast a sampled absolute position peak better than simple historical rules. An upper quantile of this peak supplies a report against which future recorded positions can be compared.

The comparison uses an hourly client-flow net-position record from one FX market maker and a weekly Treasury-bill net-position aggregate. Historical headroom and translated-path rules compete with ridge quantile regression and nearest-neighbour quantiles. Matched controls use the same completed response histories and refit schedules, so each comparison holds the target and those two design choices fixed.

The main FX regression comparison lowers mean pinball loss by 12.7%, with a HAC p-value $< .001$. Breaches nevertheless become more frequent. The original Treasury regression instead increases mean loss by 50.8%; its primary p-value is 0.008. Short-history FX regression also deteriorates at the longer horizon. These differences motivate an exploratory sequence that separates the update schedule, history length, regression specification and penalty menu.

Our empirical contribution has three parts. First, we compare established forecasts of a two-sided sampled peak using matched completed-label histories, extending the endpoint question of Wisniewski et al. (2020) (Tables 3–4). Second, we find that restrictive regularization explains much of the Treasury regression's deterioration: a shared wider penalty menu brings its main excess loss down to 3.4%, close to the intercept-only reference. The additional weekly panel shows substantial remaining losses. At the long FX history, the richer predictor set retains a 10.9% gain over the

four-predictor procedure under the same menu (Section 6.3); this compares tuning procedures and does not isolate individual lag or calendar effects. Third, we connect the scoring perspective of Gneiting (2011); Ehm et al. (2016) to joint error-amount and breach evidence, showing that the mean-score ranking need not persist across all consistent scores (Table 8 and Appendix A.6).

The comparison is retrospective, and Section 9 describes its information, statistical and decision limits. The paper proceeds from related work through the target, methods and evaluation to results, robustness and implications.

2 Related literature

Inventory research provides the economic motivation for position monitoring. Ho and Stoll (1981) model dealer quotes under inventory uncertainty, and Lyons (1995) studies inventory and information in FX trading. Duffie et al. (2023) connects Treasury-market functioning to dealer capacity. Our observed net-position series supplies an input to this motivation, while the forecasting exercise evaluates passive reports.

Financial path extrema and liquidity buffers provide the closest target precedents. MaxVaR measures the worst intermediate portfolio-value loss (Boudoukh et al., 2004). Petrunia et al. (2019) studies the maximum end-of-day net debit across payment-system participants, and van den Hoogen (2017) models a one-sided maximum cumulative payment deficit. Arjani et al. (2020) forecasts a payment-demand/resource ratio. Table 1 distinguishes these targets from the absolute peak of a single recorded position series.

Conditional quantile forecasting and its evaluation use mature tools. Regression quantiles minimize asymmetric absolute loss (Koenker and Bassett, 1978); CAViaR models time-varying conditional quantiles (Engle and Manganelli, 2004). Volatility-updated historical simulation is an established alternative for financial scenario construction (Hull and White, 1998; Barone-Adesi et al., 1999). Wisniewski et al. (2020) already applies conditional methods and lag/calendar predictors to endpoint intervals in the FX source family. Gneiting (2011) formalizes consistent quantile scores, and Ehm et al. (2016) represents their comparison through elementary scores. We use these established ideas for a common peak response.

Table 1: Closest studies and the present comparison.

Study	Target	Data	Method	Evaluation
Boudoukh et al. (2004)	Worst intermediate portfolio-value loss	Modeled portfolio paths	First-passage formula; discrete simulation	MaxVaR/VaR ratios and parameter illustrations
Petrunia et al. (2019)	Largest daily endpoint debit across ACSS participants	ACSS settlement balances	GEV blocks; GPD exceedances	Quantile diagnostics; monthly collateral shortfalls
van den Hoogen (2017)	Maximum cumulative negative payment balance	Confidential bank transactions	Modeled transaction paths; EVT	UC and independence procedures; outcomes confidential
Wisniewski et al. (2020)	Next-hour signed-position interval	Same FX source family	Quantile/conformal intervals; KNN and other regressions	Two endpoint pinball scores; interval and coverage diagnostics
This paper	Two-sided absolute sampled peak quantile	FX record; Treasury aggregate	Historical rules; QR/KNN on matched histories	Pinball, paired DM, U/V, breaches and coverage diagnostics

The FX relationship concerns the source family. The public thesis specifies backtests but withholds its application outcomes.

Relative to this literature, the comparison changes the response from an endpoint interval, directional cash deficit or portfolio loss to a two-sided sampled position peak. Matching completed-label pools and update schedules reveals that conditional accuracy depends on both the information used

and the available shrinkage: the wider menu approaches a historical headroom reference in Treasury while preserving the long-history FX gain. Joint error-amount, breach and elementary-score comparisons qualify that gain. The FX source family is that of Wisniewski et al. (2020); exact equality with their input file remains unverified.

3 Data and target

The FX record contains hourly client-flow net positions for one market maker across five liquid currency pairs, as described by Wisniewski et al. (2020). We use the hourly member of the Net Position Time Series Dataset, version two (davidlindsay1979, n.d.). It contains 18,576 observations. Adjacent observations are usually an hour apart, with weekend-sized gaps retained in their recorded order. The signed scalar is denominated in source-reported USD.

The Treasury input is New York Fed Primary Dealer Statistics series PDPOSGS-B, the aggregate Treasury-bill long position less the short position (New York Fed, 2026). The weekly Wednesday-close series contains 706 observations, denominated in USD millions. FR 2004A instructions describe market/fair-value positions on a trade-date basis, including eligible forward and when-issued outright positions, with specified settlement-date treatment for buybacks (Board of Governors, 2013, 2024). Table 2 gives the chronological splits in calendar dates.

Table 2: Source-specific chronological budgets. Dates are issue dates; all horizons share the same evaluation issues.

Source	Full record	Validation	Evaluation	W / d
FX	2014-02-05 to 2017-01-24	2015-10-08 to 2015-12-30	2016-01-01 to 2017-01-23	720 or 2880 / 24
Treasury	2013-04-03 to 2026-10-07	2019-07-24 to 2020-07-15	2021-01-06 to 2026-04-22	52 or 260 / 13

FX has 1,440 validation and 6,625 evaluation issues; Treasury has 52 and 277. The longest validation response completes before evaluation starts. H counts records, preserving recorded FX gaps.

The target includes the known issue position. Let N_t be the signed recorded position and let H count subsequent source records. Define

$$B_t = |N_t|, \qquad P_{t,H} = \max_{0 \le h \le H} |N_{t+h}|, \qquad G_{t,H} = P_{t,H} - B_t \ge 0. \tag{1}$$

Figure 1 illustrates why endpoint information can miss an intervening peak. For example, two future magnitudes can each take values zero or one with equal probability. Perfect positive dependence gives maximum one with probability one half; perfect negative dependence gives maximum one with probability one. The identical marginal distributions thus permit different peak distributions.

The information set enforces label completion before fitting. Write $\kappa(t)$ for the latest scheduled refit and W for the number of completed training origins. At segment start t_0 and refit interval d,

$$\kappa(t) = t_0 + d \left\lfloor \frac{t - t_0}{d} \right\rfloor, \qquad \mathcal{T}_{\kappa,H,W} = \{\kappa - H - W + 1, \ldots, \kappa - H\}. \tag{2}$$

Every training response ends by $\kappa(t)$. Fitted transforms and coefficients stay fixed between refits, while current query predictors and B_t update at issue. Earlier evaluation outcomes enter training only after completion. Validation and evaluation restart their schedules at their respective segment starts.

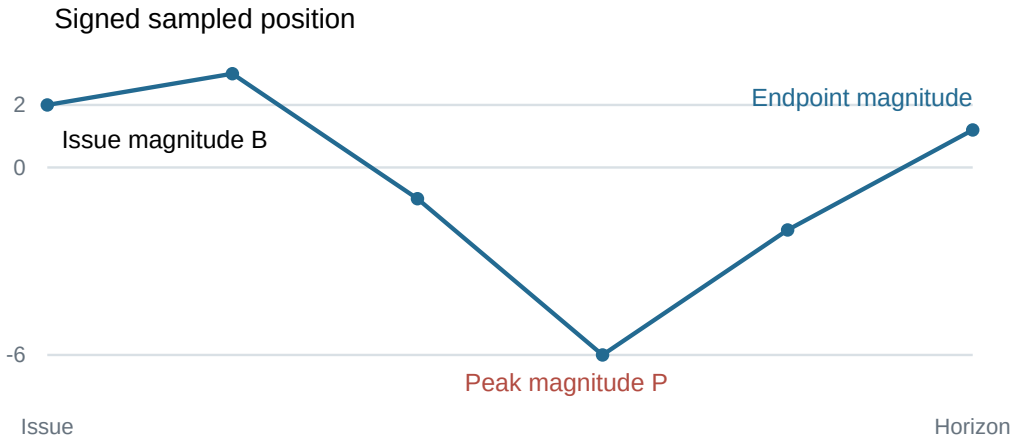


Figure 1: An artificial sampled path distinguishes the issue magnitude, the holding-interval peak and the endpoint magnitude. The illustration supplies intuition; it contains no financial observations.

Notation used in the comparison and appendices.

$N_t; B_t; P_{t,H}; G_{t,H}$	Signed position; issue magnitude; peak; headroom
$H; W; d; \kappa(t)$	Record horizon; completed-label count; refit interval; latest refit
$s_\kappa; z_{i,\kappa}; \lambda$	Response scale; standardized predictors; ridge penalty
$\ell_i; u_i; T_{i,h}^{\text{raw}}$	Lower/upper path increments; raw translated path
$R_{\text{abs}}; R_{\text{signed}}$	Maximum absolute-anchor/signed-anchor distances
$U; V; \omega$	Mean underprediction; mean overprediction; their weight ratio
$C^{\text{floor}}; \mathcal{P}_\lambda$	Issue-floor report; penalized training objective

4 Forecasting methods

4.1 Historical reports and matched controls

The historical rules differ in how they anchor completed episodes to the current issue. For each episode i, let $\ell_i = \min_h(N_{i+h} - N_i)$ and $u_i = \max_h(N_{i+h} - N_i)$, including $h = 0$. At query $x = N_t$, its raw translated peak is $T_i(x) = \max\{x + u_i, -x - \ell_i\}$. For a selected set A of size m, Q_j denotes its duplicate-preserving order statistic at $j = \lceil (m+1)\tau \rceil$, with $\tau = .95$. The three reports are

$$\begin{aligned} C_t^F &= B_t + Q_j\{G_{i,H} : i \in A\}, \\ C_t^S &= B_t + Q_j\{T_i(N_t) - B_t : i \in A\}, \\ C_t^K &= \max\{B_t, Q_j\{P_{i,H} : i \in A\}\}. \end{aligned} \tag{3}$$

Rolling controls use the most recent completed episodes at every issue. Frozen controls use $A = \mathcal{T}_{\kappa,H,W}$, matching the conditional fitting pool and update schedule. The query anchor still updates in a frozen control. Appendix A gives the finite-array relationships among these constructions.

4.2 Predictors and training transforms

The original specification contains 33 predictors: the current signed level and preceding lagged levels, current magnitude, two past increment RMS measures, recorded hour and weekday sine/cosine pairs, and a gap indicator. The level sequence spans lags zero through 24. RMS uses short and long past-only spans, defined by

$$v_L(i) = \left\{ L^{-1} \sum_{q=i-L+1}^{i} (N_q - N_{q-1})^2 \right\}^{1/2}. \tag{4}$$

The spans are daily and weekly record counts in FX, and six and 26 records in Treasury. All transforms use completed training rows. With training mean $\mu_{\kappa j}$ and population standard deviation $\sigma_{\kappa j}$,

$$z_{i,\kappa j} = \frac{x_{ij} - \mu_{\kappa j}}{\sigma_{\kappa j}}, \qquad s_\kappa = \max\left\{ W^{-1} \sum_{i \in \mathcal{T}} G_{i,H},\ 10^{-6} W^{-1} \sum_{i \in \mathcal{T}} |N_i|,\ 10^{-12} \right\}. \tag{5}$$

An exactly constant predictor is centered at its repeated value and assigned scale one. Treasury's calendar and gap coordinates are constant under its weekly sampling. Matching label histories permits different past-only predictor lookbacks; the longer FX RMS reaches before the oldest training origin.

4.3 Quantile regression and nearest neighbours

Ridge quantile regression fits normalized headroom with an unpenalized intercept. For $\rho_\tau(e) = \tau(e)_+ + (1-\tau)(-e)_+$, it solves

$$\min_{\beta_0, \beta} \frac{1}{W} \sum_{i \in \mathcal{T}} \rho_\tau\left(\frac{G_{i,H}}{s_\kappa} - \beta_0 - z_{i,\kappa}^\mathsf{T} \beta \right) + \frac{\lambda}{2} \|\beta\|_2^2, \tag{6}$$

and issues $C_t^{QR} = B_t + \max\{0, s_\kappa(\widehat{\beta}_0 + z_{t,\kappa}^\mathsf{T} \widehat{\beta})\}$. The original penalty menu is $\{.001, .01, .1\}$. Projection acts on issued reports; the training objective uses the affine headroom prediction. Appendix A.5 derives the equivalent peak parameterization and its shifted penalty.

As $\lambda \to \infty$, penalized slopes vanish and the unpenalized intercept approaches the set of empirical headroom-quantile minimizers. Write $I_t = B_t + G_{(\lceil W\tau \rceil)}$ for its lower-endpoint reference on the frozen pool. If $W\tau$ is integral, the entire interval between the adjacent order statistics minimizes the intercept-only objective. Thus I is the unique empirical quantile at the shorter Treasury history, while the longer Treasury and both FX histories admit an interval with endpoints I and corrected-rank F (possibly coincident). The two rank conventions need not issue the same report.

Nearest-neighbour quantiles select standardized predictor rows by a group-balanced Euclidean distance and resolve equal distances by recency. Before computing distance, each of the lagged-level coordinates is divided by five; each coordinate in the RMS, hour and weekday pairs is divided by $\sqrt{2}$. Magnitude and gap retain their standardized scales. Plain KNN applies the corrected order statistic to selected headrooms. Scaled KNN first divides each selected headroom by a short-run RMS scale and multiplies the resulting quantile by the current scale. The RMS floor is $10^{-6} s_\kappa$. The supplement records the fixed neighbourhood menus for each source and history. At $k = W$, plain KNN equals frozen F because membership is the full pool.

Validation selects the lowest native-unit mean pinball loss and resolves an exact parameter tie toward the larger candidate. Every declared candidate must complete for the cell to be eligible. The full validation and evaluation inventories retain all numerical failures and adverse outcomes. Appendix C gives the reproduction route and numerical acceptance scope.

5 Evaluation design

The two source-specific main comparisons were declared before their conditional evaluation runs. We call these pre-specified main comparisons the primaries: FX regression at the long history versus the original rolling short-history F, and Treasury regression versus its own long-history frozen F. Their horizons are both six records. The longer-horizon FX comparison against the original rolling S is a legacy-reference sensitivity. The original designs, menus and selected parameters are publicly recorded in the protocol archive. The FX evaluation archive had already been inspected before specifying the conditional extension. AR replay, rank and loss-weight diagnostics, statistical inference and the regression specification analysis are later exploratory additions.

Mean score and breach frequency answer complementary questions. For report C_t, define

$$U = \frac{1}{M}\sum_t (P_{t,H} - C_t)_+, \quad V = \frac{1}{M}\sum_t (C_t - P_{t,H})_+, \quad \overline{L} = .95U + .05V. \tag{7}$$

A strict breach is $P_{t,H} > C_t$; equality is a nonbreach. Both error amounts average over all issues. Mean headroom $C_t - B_t$ measures a different quantity from V. All scores use the original source units, and each percentage comparison uses its own reference mean.

Paired loss differentials support descriptive unconditional predictive-ability comparisons. For $d_t = L_t^A - L_t^B$, we report $\bar{d}/\sqrt{\widehat{\Omega}/M}$ using Bartlett Newey–West HAC (Diebold and Mariano, 1995; Newey and West, 1987). The bandwidth is $L = \max\{H, \lfloor 4(M/100)^{2/9} \rfloor\}$. It is ten lags for short-horizon FX and equals the horizon in the other cells. We also report twice-bandwidth sensitivity and the Harvey–Leybourne–Newbold correction (Harvey et al., 1997). That correction originates in mean-squared-error comparisons; its use with pinball loss is approximate. The fixed-window interpretation follows unconditional predictive ability in Giacomini and White (2006). Serial regularity and a suitable finite long-run variance remain assumptions of these reference distributions.

We report the two previously declared source-specific primaries with unadjusted HAC p-values and a Bonferroni sensitivity across those two comparisons. Holm (1979)'s step-down procedure covers the remaining 71 original matched conditional-versus-frozen-$F/S/K$ contrasts. Tables 3–4 identify the Treasury primary and otherwise display the applicable adjusted own-F comparisons; historical S/K contrasts and the FX cross-window primary lie outside that matched family. This primary-versus-supplementary reporting division was chosen after inspecting results, and the former 72-member adjustment remains in the archive. Together with the selection and inspection history, this makes the p-values exploratory evidence rather than confirmatory control for the entire research process.

Coverage diagnostics use the single grid $t_0 + j(H+1)$, anchored once at evaluation start. The retained target windows share no recorded observations. Training histories and wider serial dependence can still overlap. We report full-grid Kupiec proportion-of-failures tests (Kupiec, 1995), plus Christoffersen independence and conditional-coverage likelihood ratios conditional on the first retained hit (Christoffersen, 1998). The latter uses the subsequent hits consistently in both its null and alternative likelihoods. Treasury's two grids contain only 40 and twelve observations, so asymptotic reference p-values have limited precision and power. The exact nominal-exceedance null also differs from an upper-bound null when the peak distribution has an atom at its known floor.

6 Results

6.1 FX: a long-history gain and a short-history reversal

Long-history regression improves the six-step FX score. Its primary reduction is 12.7% against rolling F, with a descriptive HAC statistic of approximately -5.2. Against its own frozen F, the reduction is 14.8%. The matched statistic is approximately -5.9. Table 3 gives the complete original conditional menu and the historical controls. The long-history gain coexists with more breaches, a distinction developed below.

Short-history regression has a much weaker six-step result. Its matched score reduction is 2.12%, with an unadjusted p-value of 0.54. At the longer horizon, the same short-history procedure increases mean loss by about 32%. Extending history improves the regression comparison, whereas extending frozen F slightly raises its short-horizon loss. This contrast motivates the ordered specification analysis.

The longer-horizon legacy gain is small and statistically imprecise. Regression improves mean loss by 1.83% against the original rolling S, with an unadjusted p-value of 0.84. The partial final year contributes about 71% of its aggregate improvement. By comparison, that partial year contributes

Table 3: FX original reports, matched by horizon and history. Loss is in source-reported USD millions.

H	W	Report	Mean loss	Δ_F (%)	Breach (%)	p
6	720	F	0.443	+0.000	5.011	–
6	720	S	0.451	+1.843	3.849	–
6	720	K	0.932	+110.664	7.758	–
6	720	QR	0.433	-2.117	8.015	1.000
6	720	KNN	0.436	-1.394	4.815	1.000
6	720	Scaled KNN	0.466	+5.349	4.060	1.000
6	2880	F	0.453	+0.000	5.434	–
6	2880	S	0.451	-0.418	4.045	–
6	2880	K	0.972	+114.751	8.091	–
6	2880	QR	0.386	-14.767	5.675	< 0.001
6	2880	KNN	0.411	-9.302	4.815	0.016
6	2880	Scaled KNN	0.426	-5.795	4.121	1.000
24	720	F	0.787	+0.000	6.445	–
24	720	S	0.755	-4.078	4.362	–
24	720	K	1.206	+53.135	11.049	–
24	720	QR	1.035	+31.504	13.947	0.066
24	720	KNN	0.836	+6.197	7.592	1.000
24	720	Scaled KNN	0.897	+13.886	6.642	1.000
24	2880	F	0.763	+0.000	5.389	–
24	2880	S	0.759	-0.525	4.408	–
24	2880	K	1.240	+62.440	8.936	–
24	2880	QR	0.735	-3.758	8.015	1.000
24	2880	KNN	0.770	+0.837	6.883	1.000
24	2880	Scaled KNN	0.848	+11.085	5.962	1.000

Every relative change uses its own frozen F. *Primary, unadjusted HAC p-value (Treasury QR at the main setting). Other conditional p-values use Holm over the remaining 71 matched conditional-versus-$F/S/K$ contrasts. The FX cross-window primary is reported separately. This reporting division was chosen after results; the original 72-member adjustment remains archived. Dashes mark self-references, historical contrasts or the exact full-pool KNN identity. Identical-loss cells receive input one for conservative bookkeeping. Values are rounded only for display.

about 8.6% of the six-step primary improvement. The annual and sparse-grid outputs therefore help distinguish stable directions within this archive from gains concentrated in a short subdivision.

6.2 Treasury: the original regression loses

The original Treasury regression is less accurate than matched historical headroom. Its main score increases by 50.8%. Strict breaches rise from 28 to 54. Its primary HAC p-value is 0.008 (0.015 with the two-primary Bonferroni sensitivity). All original Treasury regression and Scaled KNN cells lose to their own frozen F controls. Each accepted fit passed the numerical conditions, so these losses are forecasting outcomes.

Table 4: Treasury original reports, matched by horizon and history. Loss is in source-reported USD millions.

H	W	Report	Mean loss	Δ_F (%)	Breach (%)	p
6	52	F	2779.054	+0.000	3.971	–
6	52	S	2786.620	+0.272	3.971	–
6	52	K	2574.098	-7.375	10.108	–
6	52	QR	5095.519	+83.354	30.325	0.112
6	52	KNN	2779.054	+0.000	3.971	–
6	52	Scaled KNN	4208.293	+51.429	3.249	0.013
6	260	F	2758.665	+0.000	10.108	–
6	260	S	2707.244	-1.864	10.108	–
6	260	K	2287.758	-17.070	10.469	–
6	260	QR	4159.673	+50.786	19.495	0.008*
6	260	KNN	2663.561	-3.447	10.108	1.000
6	260	Scaled KNN	3127.173	+13.358	8.303	1.000
24	52	F	3715.693	+0.000	11.913	–
24	52	S	3652.379	-1.704	11.913	–
24	52	K	5017.741	+35.042	25.271	–
24	52	QR	12165.574	+227.411	47.653	0.983
24	52	KNN	3715.693	+0.000	11.913	–
24	52	Scaled KNN	5625.667	+51.403	11.552	0.226
24	260	F	2480.306	+0.000	4.332	–
24	260	S	2478.871	-0.058	4.332	–
24	260	K	2301.476	-7.210	24.188	–
24	260	QR	5204.760	+109.843	30.325	1.000
24	260	KNN	2257.963	-8.964	4.693	1.000
24	260	Scaled KNN	4137.264	+66.805	2.888	< 0.001

Every relative change uses its own frozen F. *Primary, unadjusted HAC p-value (Treasury QR at the main setting). Other conditional p-values use Holm over the remaining 71 matched conditional-versus-$F/S/K$ contrasts. The FX cross-window primary is reported separately. This reporting division was chosen after results; the original 72-member adjustment remains archived. Dashes mark self-references, historical contrasts or the exact full-pool KNN identity. Identical-loss cells receive input one for conservative bookkeeping. Values are rounded only for display.

Plain KNN at the short Treasury history reproduces F. Validation selects the entire eligible pool at both horizons. Its longer-history, longer-horizon score gain reverses on the twelve-point grid: full-sample improvement becomes sparse-grid deterioration. Both reports have two strict breaches on that grid. These outcomes preserve the distinction between a complete-sample mean, a small sensitivity subset and an independent validation sample.

Coverage diagnostics reinforce the original Treasury regression's adverse breach record. Table 5 gives the nonoverlapping-grid results beside the FX comparison. A failure to reject for the historical reference has limited evidential strength in this small grid.

Table 5: Six-step coverage diagnostics on the fixed nonoverlap grid.

Report	n	Breaches	Rate (%)	p_{POF}	p_{IND}	p_{CC}
FX QR	947	50	5.280	0.695	0.388	0.636
Treasury QR	40	7	17.500	0.004	0.775	0.014
Treasury F	40	4	10.000	0.199	0.338	0.262

POF uses every retained hit. IND and CC condition on the first hit; their likelihoods therefore use the subsequent transitions consistently. Values are asymptotic reference p-values for equality/independence nulls. Complete grids and the preserved earlier initialization convention are in the inference supplement.

6.3 Ordered specification analysis

The exploratory sequence asks which procedural changes accompany the score differences. We start from rolling F, freeze the update schedule, extend the completed-label history, change to a four-predictor regression, and then change to the original full predictor set. The parsimonious regression retains current signed position, current magnitude and the two past RMS measures. Figure 2 displays successive differences in loss units normalized by a common starting reference; its increments therefore sum to the total.

The original sequence combines two changes at its last step. The four-predictor fit used the extended penalty menu $\{.001, .01, .1, 1, 10\}$, while the published full specification used the smaller original menu. Every Treasury parsimonious cell selected the strongest available penalty. E3 therefore fits the full specification using the same extended menu and preserves the original splits, refit schedule, tie rule and numerical conditions. It is a post-result comparison and leaves both published primaries in place.

A wider penalty menu largely removes the original Treasury deterioration. At the main Treasury setting, the full regression's excess loss relative to F falls from 50.8% to 3.43%. Its penalty changes from 0.001 to one. Under the common menu, the full predictor set increases loss by 1.24% relative to the parsimonious regression, with a descriptive p-value of 0.685. The small difference leaves substantial uncertainty; it is not an equivalence result. Across the Treasury cells, the full model's matched-F changes now range from a modest improvement to a modest deterioration.

The FX gain persists under the wider menu. Every FX selection and issued report agrees with the original full specification. At the long history, changing from the parsimonious to the full predictor set reduces loss by 10.9%, with an E3-adjusted HAC p-value below 0.001. Changing from frozen F to the parsimonious model yields a smaller reduction; that step has an unadjusted p-value of 0.010 and an E3-adjusted value of 0.251. Table 7 retains every step, including the update and history changes.

The revised interpretation concerns regularization and specification jointly. Stronger shrinkage is useful for the weekly Treasury histories, while the long-history FX specification retains its gain. The shared-menu comparison permits each feature set to select its own penalty. Its order-dependent differences support a conditional comparison of procedures, rather than a causal attribution to a particular predictor or a universal dimension rule. The intercept-only reference helps locate the remaining Treasury differences. In the displayed Treasury order, I changes loss by -2.0%, $+3.2\%$, $+6.2\%$, $+2.4\%$ relative to F. The expanded full regression selects the menu's upper endpoint in 3 of 4 Treasury cells. At the main Treasury setting, expanded QR has 32 strict breaches, compared with 28 for F and 54 for original QR. Stronger shrinkage therefore narrows the score deterioration without restoring the historical breach count.

An additional panel reuses 12 previously archived New York Fed position series on the same economic-date grid, with all 48 horizon–history cells retained (Figure 3). Expanding the full regression's menu reduces the median excess loss relative to F from $+92.6\%$ to $+9.2\%$, but it still loses in 34 cells, compared with 45 originally. The committed pre-run predictions yield P1: 30/45 against a threshold of 40; P2: 24/48 against a threshold of 36; P3: 33/48 against a threshold of 25.

Table 6: Penalty-menu expansion and the intercept-only reference. Loss changes relative to frozen F (percent); the final column counts strict breaches.

Source	H	W	λ (33+)	Original 33	Four+	33+	I	Breaches (33+)
FX	6	720	0.01	-2.12	+1.48	-2.12	-0.15	531
FX	6	2880	0.01	-14.77	-4.29	-14.77	+0.01	376
FX	24	720	0.01	+31.50	+16.51	+31.50	-0.00	924
FX	24	2880	0.001	-3.76	-2.89	-3.76	+0.03	531
Treasury	6	52	10	+83.35	-2.59	-3.24	-1.99	14
Treasury	6	260	1	+50.79	+2.17	+3.43	+3.24	32
Treasury	24	52	10	+227.41	+5.95	+4.34	+6.21	39
Treasury	24	260	10	+109.84	+0.11	-0.19	+2.35	17

Plus denotes the common menu $\{.001, .01, .1, 1, 10\}$. Original 33 retains its smaller menu. I uses the ordinary empirical headroom quantile at $\lceil .95W \rceil$; it is a $\lambda \to \infty$ reference, with a minimizer interval when $.95W$ is integral. F instead uses $\lceil .95(W+1) \rceil$. Each predictor set selects its own penalty by the same validation rule. All eight cells and all 40 candidates completed. Common-menu paired tests, the separate 28-comparison E3 adjustment, HLN and doubled-bandwidth results remain in the supplement.

Table 7: Ordered six-step changes ending at the original main comparisons.

Source	Change	Step (%)	Initial (pp)	p_{DM}	p_{HLN}
FX	Rolling F_{720}→Frozen F	+0.190	+0.190	0.297	0.297
FX	Frozen F_{720}→Frozen F_{2880}	+2.277	+2.281	0.143	0.143
FX	Frozen F→Four+	-4.289	-4.395	0.010	0.010
FX	Four+→33+	-10.947	-10.736	$< .001$	$< .001$
FX	33+→Original 33	+0.000	+0.000	–	–
Treasury	Frozen F→Four+	+2.166	+2.166	0.178	0.188
Treasury	Four+→33+	+1.242	+1.269	0.685	0.691
Treasury	33+→Original 33	+45.779	+47.352	0.006	0.007

Step percentages use the preceding row's loss and compound. Initial percentage points use the chain's common starting loss and sum to the original total. The last Treasury step restricts the ridge menu again, separating that change from the common-menu predictor comparison. Dashes denote identical losses. Tests are exploratory and unadjusted here; the declared E3 family and twice-bandwidth checks are in the supplement.

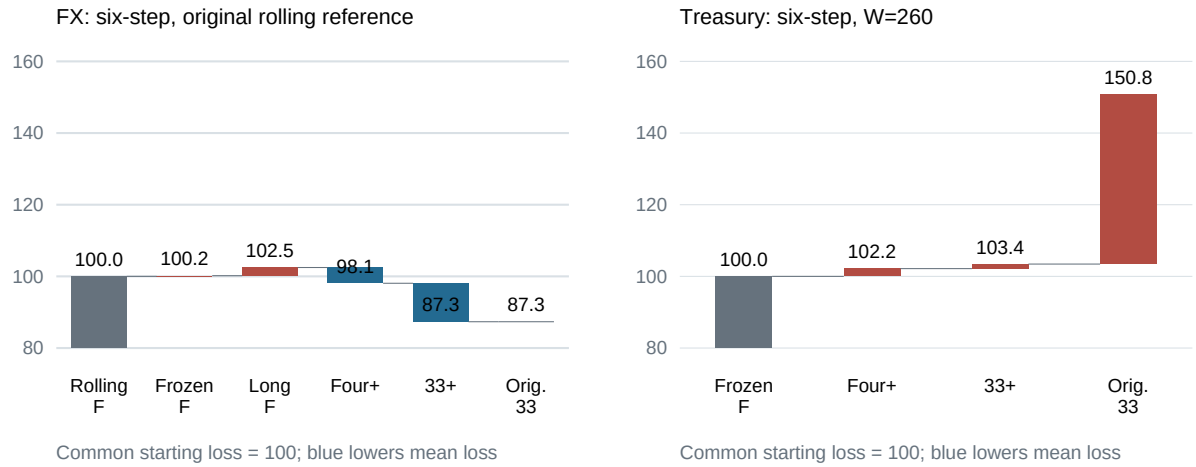


Figure 2: Ordered loss changes for the six-step comparisons. Each bar changes the preceding procedure; the common-baseline vertical scale makes the increments additive. The original full QR and the equal-menu E3 report are both retained. Corresponding paired HAC tests appear in the decomposition table. The chosen order defines the result; it is a procedure comparison rather than a Shapley allocation or a causal feature effect.

P1 and P2 are unsupported; P3 is supported. Shrinkage therefore mitigates rather than eliminates the observed weakness. The design followed inspection of the original panel results, and the series share a reporting system and overlapping dealer positions; this is an exploratory extension, not independent replication.

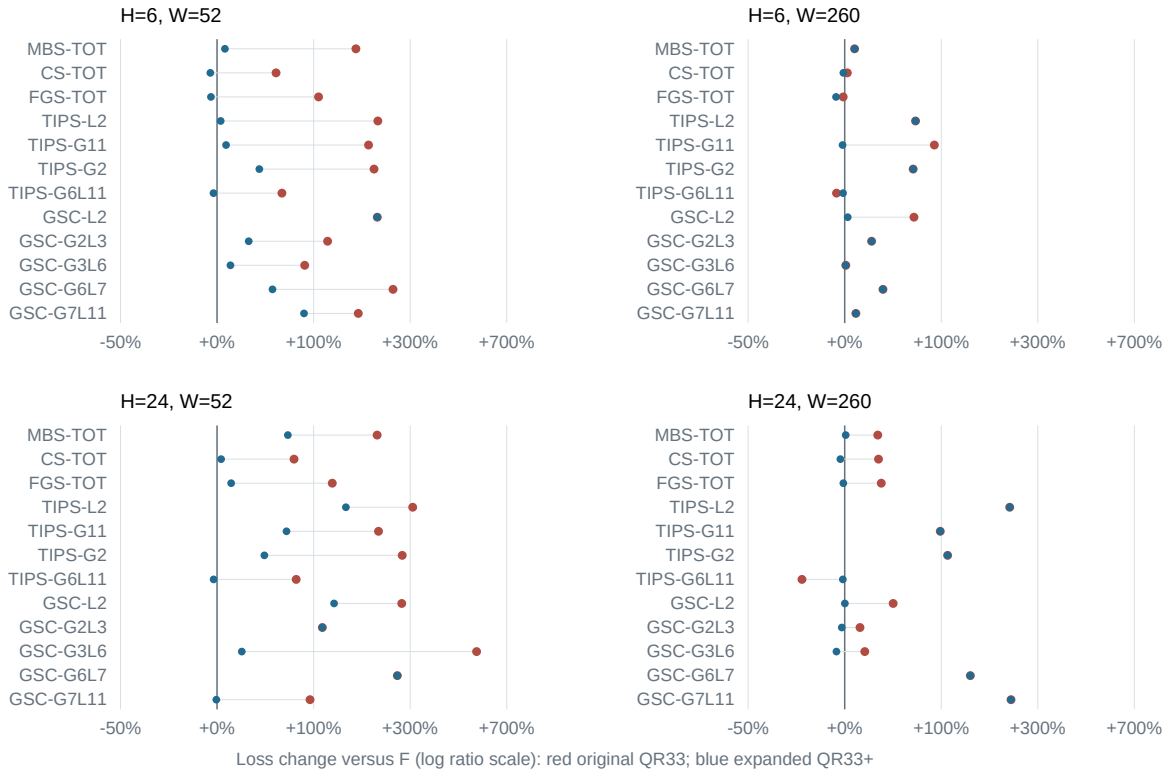


Figure 3: All archived panel cells, each comparing original and expanded full regression against its own F. The axis shows relative loss on a logarithmic ratio scale; zero is the historical reference. Row labels omit the common PDPOS prefix. Complete native-unit losses, selected penalties, I and breach counts are retained in the E5 table.

7 Robustness and score interpretation

AR residual-block replay shows that family choice matters. At each original refit, OLS estimates an intercept and slope on unique transitions from the union of completed windows. Ordered residuals are replayed from the current anchor; Appendix A.4 gives the exact path identity. AR improves on regression in every Treasury cell and loses in most FX cells. Against frozen translated paths, its score improves in five cells, while breaches rise in every cell. This is a mean-filtered comparison with a separate residual-representativeness assumption.

Rank sensitivity changes the finite-sample convention. The original historical rank is $\lceil (m+1)\tau \rceil$; ordinary empirical quantiles use $\lceil m\tau \rceil$, with an interval of minimizers at integral $m\tau$. The short Treasury full-pool case has distinct ranks under those rules. The rank-ablation supplement retains all contrasts, including reversals, and the full-pool KNN identity persists whenever both reports share their rank convention. Results concern the issued reports under the stated convention.

Loss reweighting distinguishes a robust linear gain from a narrow tradeoff. For each fixed issued report, write $J_\omega = \omega U + V$, so the original quantile score corresponds to $\omega = 19$ up to a positive factor. Table 8 shows that the short-horizon FX primary reduces both components. Its breach count nevertheless increases from 332 to 376. The longer-horizon legacy gain instead exchanges higher underprediction for lower overprediction, and reverses above a weight ratio of 20.6. These

statements concern constant linear weights on the archived reports. The elementary-score comparisons in Appendix A.6 cross for both primaries, ruling out empirical dominance over the full class of consistent quantile scores.

Table 8: Error components and strict breaches for the original named comparisons.

Comparison	ΔU	ΔV	Crossing ω	Breaches A/B
FX six-step primary	-0.010	-0.921	–	376/332
FX six-step matched	-0.030	-0.776	–	376/360
FX longer-horizon legacy	+0.168	-3.463	20.631	531/287
Treasury primary	+1736.223	-4968.078	2.861	54/28

A is regression and B its named reference. Amounts are in source-reported USD millions, with distinct source measurement processes. Negative component differences favor A. A missing crossing means both components have the same sign. The weights describe hypothetical fixed-report preferences.

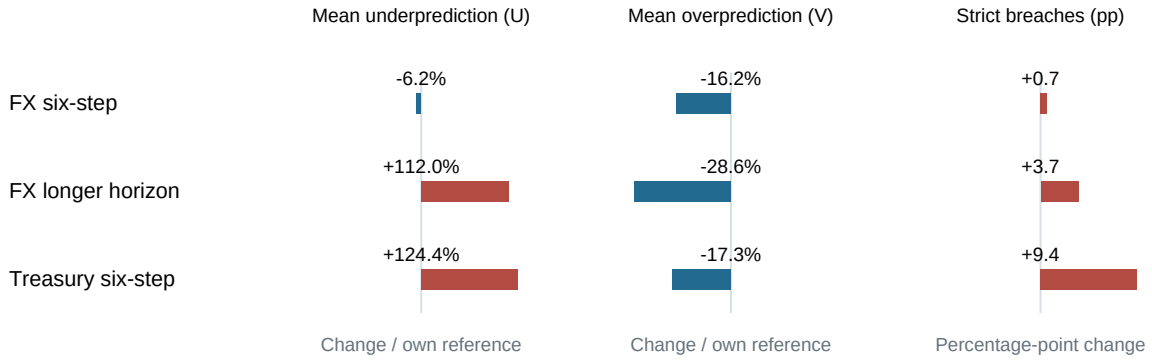


Figure 4: Mean underprediction (U), mean overprediction (V) and strict-breach changes for the named original comparisons. Each amount panel scales its own reference to 100; the breach panel shows percentage-point changes. FX and Treasury amounts describe distinct financial objects.

Horizon diagnostics show that realized peaks and fitted reports have different ordering properties. Realized peaks increase weakly with horizon on every common issue. Separately fitted quantile reports can cross; the original long-history Treasury regression crosses on about 29% of issues. A post-result adjustment that raises a longer report can also worsen its score, as the appendix illustrates. Table 9 summarizes these checks while the complete archived menus remain available in the supplement.

Table 9: Compact robustness summary; the complete original outputs are retained.

Check	Finding	Interpretation
AR residual replay	Lower loss than translated paths in five cells; more breaches in every cell	Conditional-family rankings differ
Rank convention	Ordinary versus corrected rank changes some finite reports; full-pool identities persist	Report construction matters
Fixed linear weights	FX short-horizon components both fall; longer-horizon crossing near the original weight	Mean score and breach count differ
Horizon order	Targets are ordered; fitted reports can cross	Separate fits need an explicit reconciliation criterion

8 Implications for position monitoring

Keep a historical benchmark. Historical headroom is a demanding reference for these sampled position peaks. The weekly results show why a conditional method should be compared with a historical rule on the same completed labels and update schedule before it is used for monitoring.

Offer a genuine shrinkage option. A conditional selection menu should include an intercept-only report and the exact historical benchmark as explicit candidates. The ordinary empirical quantile need not equal corrected-rank F, so a large finite penalty or I alone does not necessarily nest F. Including F protects the minimum validation score under the same validation rule, while evaluation noninferiority remains unguaranteed.

Report error amounts and breach occasions together. Mean underprediction and overprediction describe the size of opposite forecast errors, whereas strict breaches count their occurrence. A monitoring report should display all three alongside the stipulated score, since a lower score alone does not establish fewer limit exceedances or lower observed funding cost.

9 Discussion and limitations

Inspection and inference. The FX evaluation period had been inspected before the conditional methods were specified, so it is not a clean untouched test. Later statistical inference and the specification sequence describe that same archive. HAC regularity, fixed-window predictive-ability premises and the effective sample information behind overlapping labels require care; a nonoverlap grid does not make the observations independent.

Measurement and information timing. Treasury is a downloaded-vintage replay indexed by economic date, not a reconstruction of historical public releases. The aggregate can hide offsetting dealer positions and includes valuation and composition effects. Both sources sample a position process and omit intrarecord excursions; the FX client-flow record is not a complete hedged dealer book.

Coverage and specification. These procedures have no established conditional-coverage guarantee for these records. Floor atoms distinguish an exact nominal-exceedance null from the upper-bound implication of a quantile, while Treasury's sparse grids make likelihood-ratio approximations fragile. Matching the penalty menu removes one specification difference, but the ordered regression comparison cannot identify a universal or causal value of extra predictors.

Decision value. We have no observations of the trading, hedging, collateral or funding decisions needed to measure economic benefit. A lower stipulated score therefore does not establish lower realized financing cost. Active inventory-sensitive decisions would also alter the position process that a passive forecast observes.

The empirical object is a measured position magnitude; its connection to a particular inventory policy requires the corresponding decision setting.

10 Conclusion

Historical rules are demanding benchmarks for forecasts of sampled position peaks. In the hourly FX record, long-history regression lowers the six-step primary pinball loss by 12.7%, although breaches increase. In the weekly Treasury record, expanding the ridge menu reduces the original regression's excess loss from 50.8% to 3.4%, close to an intercept-only empirical-quantile reference. Across the additional weekly panel, the corresponding median excess falls from 92.6% to 9.2%, yet 34 of 48 cells still lose to historical headroom. Stronger shrinkage explains substantial deterioration without eliminating it.

These findings favor a selection menu containing both an intercept-only option and the exact historical benchmark, accompanied by joint reporting of underprediction, overprediction and breaches. Including a benchmark can protect the minimum validation score; it cannot guarantee evaluation performance. The surviving FX gain concerns a richer predictor procedure, without identifying a causal contribution from individual predictors. Neither primary uniformly dominates across all consistent quantile scores. Prior inspection, overlapping targets, downloaded-vintage Treasury observations and the panel's shared reporting system limit inference. The evidence concerns these retrospective forecasting exercises and does not establish gains from changing actual inventory or funding decisions.

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A Finite geometry and parameterization

These elementary arguments concern a common nonempty historical set J, an available rank $1 \le j \le |J|$ and real arithmetic. Write $x = N_t$, $B = |x|$, $B_i = |N_i|$, and $d_{i,h} = N_{i+h} - N_i$, including $d_{i,0} = 0$. The quantities $\ell_i = \min_h d_{i,h}$ and $u_i = \max_h d_{i,h}$ retain the notation of the main text.

A.1 Common-set projection bounds

Define $A_i = B + P_i - B_i$, $T_i = \max_h |x + d_{i,h}|$ and the floor projection $\Pi_B(v) = \max(B, v)$. Both A_i and T_i are at least B. A nondecreasing projection commutes with a finite order statistic, so $F_J = Q_j(A)$, $S_J = Q_j(T)$ and $K_J = Q_j(\Pi_B(P))$. The projection is one-Lipschitz; therefore

$$|A_i - \Pi_B(P_i)| \le |B - B_i|, \qquad |T_i - \Pi_B(P_i)| \le |x - N_i|. \tag{8}$$

The second bound follows by applying the reverse triangle inequality to every path point, then taking the maximum and applying the floor projection. Order statistics are nonexpansive in the maximum paired-row difference: if $|a_i - b_i| \le R$ for each i, then $a_i \le b_i + R$ and conversely, which implies $|Q_j(a) - Q_j(b)| \le R$. Consequently,

$$|F_J - K_J| \le R_{\text{abs}}, \quad |S_J - K_J| \le R_{\text{signed}}, \quad |F_J - S_J| \le R_{\text{abs}} + R_{\text{signed}}, \tag{9}$$

where $R_{\text{abs}} = \max_{i \in J} |B - B_i|$ and $R_{\text{signed}} = \max_{i \in J} |x - N_i|$. Among size-$m$ subsets, choosing the smallest signed distances minimizes their largest distance, since every other subset contains a distance at least as large as the mth ordered distance. This minimax statement concerns the signed radius. For any realized response, pinball loss is $\max(\tau, 1 - \tau)$-Lipschitz in its report, so the report bounds also bound absolute score differences. They supply no direction of score improvement and require common rows and rank.

A.2 Envelope plateaus and signed anchors

Put $v_i = -\ell_i \ge 0$. The headroom of an increment envelope at anchor z is

$$g_i(z) = \max(z + u_i, -z + v_i) - |z| = \begin{cases} \max(u_i, v_i - 2z), & z \ge 0, \\ \max(u_i + 2z, v_i), & z \le 0. \end{cases} \tag{10}$$

The branches agree at zero. Thus g_i is two-Lipschitz and lies between $\min(u_i, v_i)$ and $\max(u_i, v_i)$. Comparing $g_i(N_i)$ with $g_i(x)$ row by row and using order-statistic stability gives

$$|F_J - S_J| \le \max_{i \in J} \min\{2|x - N_i|, |u_i - v_i|\}. \tag{11}$$

Symmetric envelopes, $u_i = v_i$, suffice for equality of F_J and S_J. Nonnegative historical/query anchors and $u_i \ge v_i$ also suffice; the sign-reversed analogue interchanges the two envelope sides. These are sufficient conditions. The same range bound yields $F_J, S_J \le B + Q_j\{\max(u_i, v_i)\}$. If every $B_i \le B$, then $F_J \ge K_J$; if every $B_i \ge B$, the order reverses. Both follow directly by comparing A_i with $\Pi_B(P_i)$.

Equal absolute anchors can still leave a translated-path gap. Take $c > 0$, $H = 1$, $t = 721$ and the history $N_s = (-1)^{s+1}c$. The completed pool $J = \{1, \dots, 720\}$ contains equal counts of both signs and the query is c. All native peaks are c and all headrooms vanish. Raw translation gives half the peaks c and half $3c$, so rank 685 yields $F = K = c$ and $S = 3c$. Here $R_{\text{abs}} = 0$ and $R_{\text{signed}} = 2c$.

A.3 Compatible futures with opposite score rankings

Fix $d > 0$ and an available corrected rank for a pool of size n. Set $t = H + n - 1$, $N_s = d(t - s)$ through issue, and use origins $J = \{0, \dots, n-1\}$. Every historical episode declines, giving $G_i = 0$, while the current anchor is zero. Hence $F = 0$ and $S = Hd$. A future staying at zero gives peak zero and strictly favors F; a continuation $N_{t+h} = -dh$ gives peak Hd and strictly favors S. The identical observed history therefore permits either realized score ordering.

Report order can also differ across constructions. The artificial history $(0, -4, -1, 2)$ with $H = 1$ has completed origins $0, 1, 2$ and current anchor two. Their native peaks are $(4, 4, 2)$, headrooms $(4, 0, 1)$ and translated peaks $(2, 5, 5)$. At the available corrected rank three for $\tau = 3/4$, this gives $F = 6$, $S = 5$ and $K = 4$. This auxiliary quantile belongs only to the illustration.

A.4 Fixed-coefficient residual replay

Fix arbitrary c, ϕ and define residuals $\epsilon_s = N_{s+1} - c - \phi N_s$. Replaying a historical residual sequence from x gives $M_{i,0}(x) = x$ and $M_{i,h+1} = c + \phi M_{i,h} + \epsilon_{i+h}$. Subtracting the observed recursion and using induction establishes

$$M_{i,h}(x) = N_{i+h} + \phi^h (x - N_i), \qquad T^{\mathrm{raw}}_{i,h}(x) = N_{i+h} + (x - N_i). \tag{12}$$

Taking absolute maxima and using the reverse triangle inequality gives

$$\left| \max_h |T^{\mathrm{raw}}_{i,h}(x)| - \max_h |M_{i,h}(x)| \right| \le \max_{0 \le h \le H} |1 - \phi^h| \, |x - N_i|. \tag{13}$$

For $0 \le \phi \le 1$, the coefficient is $1 - \phi^H$; at $\phi = 1$ the paths coincide. The identity is algebraic and needs no residual distribution. Treating historical blocks as representative future scenarios is an additional modeling premise. The replay's time order generally requires more information than the two raw increment extrema.

Horizon order alone leaves score effects unresolved. For example, $B = 0$, both realized peaks equal one, the short report is ten, and the long report is one. Raising the long report to the short one changes its quantile loss from zero to 0.45. This separates a known response floor from an ordering imposed between fitted reports.

A.5 Peak parameterization of headroom regression

The current-magnitude predictor satisfies $B_i = \mu_{\kappa B} + \sigma_{\kappa B} z_{i,\kappa B}$ under the stated transform. Set

$$\alpha = \mu_{\kappa B} + s_\kappa \beta_0, \qquad \theta = s_\kappa \beta + \sigma_{\kappa B} e_B. \tag{14}$$

Positive homogeneity of pinball loss implies that multiplying the objective in (6) by s_κ gives

$$\mathcal{P}_\lambda(\alpha, \theta) = \frac{1}{W} \sum_{i \in \mathcal{T}} \rho_\tau(P_{i,H} - \alpha - z_{i,\kappa}^{\mathsf{T}} \theta) + \frac{\lambda}{2 s_\kappa} \|\theta - \sigma_{\kappa B} e_B\|_2^2. \tag{15}$$

The issued report is $\max\{B_t, \alpha + z_{t,\kappa}^{\mathsf{T}} \theta\}$. The equivalent peak problem thus has a shifted slope penalty. For every response $P \ge B$, raising a report below B to B weakly reduces its realized pinball loss. Translation by a known B also translates the corresponding conditional quantile. These finite relationships leave estimation and future calibration as separate questions.

A.6 Elementary quantile scores

For a threshold η, the elementary score for forecast c and outcome p is

$$S_{\eta,\tau}(c,p) = \{\mathbf{1}(p < c) - \tau\}\{\mathbf{1}(\eta < c) - \mathbf{1}(\eta < p)\}. \tag{16}$$

Ehm et al. (2016) represents quantile-consistent scores as nonnegative mixtures of these elementary scores under the appropriate regularity conditions. The empirical difference is constant between forecast/outcome breakpoints, with equality conventions checked at the breakpoints. Both named primaries take positive and negative values over the full breakpoint set. Figure 5 displays a reduced plotting grid; the sign audit uses all breakpoints. The crossing establishes absence of empirical uniform dominance over consistent scores; it supplies no population dominance test.

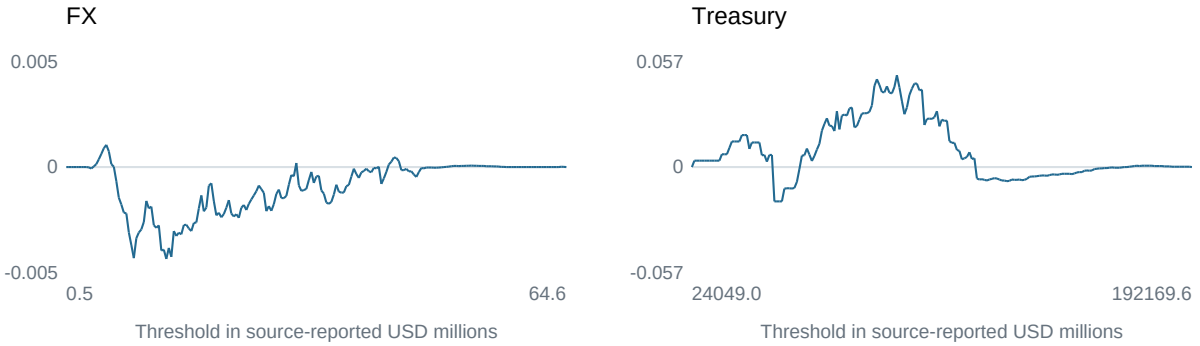


Figure 5: Elementary-score differences for the two original primaries, regression minus its declared reference. Positive and negative segments occur in each panel. Thresholds follow the respective source scales; the plotting subset preserves the audit's full-range sign conclusion.

A.7 Descriptive penalty paths

Figure 6 extends the penalty grid for the two original six-step long-history settings. The validation-selected models and original comparisons remain fixed. Evaluation curves are diagnostic and do not select a replacement penalty. An evaluation score need not improve monotonically with shrinkage, and the intercept-only optimum can be an interval rather than a unique report.

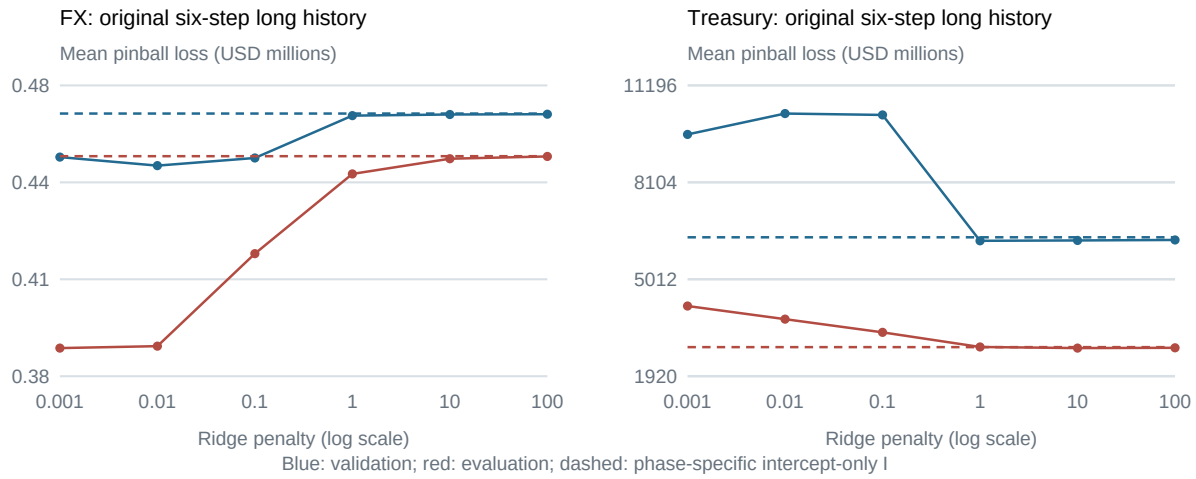


Figure 6: Descriptive validation and evaluation loss paths with phase-specific intercept-only references. The complete grid, reused and newly fitted reports, and numerical receipts are archived under E6. No point on the evaluation curve is used for selection.

B Earlier construction evidence

Table 10 retains the original historical-rule grid. R is the symmetric-increment-envelope rule and $C^{\text{floor}} = B$ is the issue-floor report. The earlier one-state retrieval study then compared recent and nearest signed-state pools. Its nearest native-peak improvement left the stronger original F/S references ahead. The complete retrieval menu, monthly subdivisions and stationary-block sensitivities remain in the original result supplement; they are historical evidence from this same record.

Table 10: Original historical construction grid; mean FX losses in USD millions.

Setting	F	S	R	K	C^{floor}	$F - S$
main	0.442	0.451	0.528	0.926	1.814	-0.009
h12	0.593	0.605	0.690	1.059	2.951	-0.012
h24	0.777	0.748	0.803	1.198	4.631	+0.029
n480	0.450	0.461	0.533	0.885	1.814	-0.011
n1440	0.444	0.448	0.527	0.932	1.814	-0.004
tau90	0.660	0.659	0.763	1.477	1.718	+0.000
tau99	0.155	0.159	0.175	0.289	1.890	-0.004

Each sensitivity changes one factor from the original six-step setting. All rows retain the same evaluation issues; different score preferences are separate tasks. The displayed +0.000 in the tau90 row is positive before rounding.

C Reproducibility and technical supplement

The repository separates frozen original results, new exploratory results and document sources. The original working paper remains available beside this revision. The revision guide maps each current table and figure to its source file, records moved material, and provides the exact compilation command. Conventional bibliography entries replace reading-log annotations; the underlying source-evidence notes preserve consulted versions and ranges.

Exact recomputation requires the pinned source snapshots, model specification and code. The Treasury snapshot is archived with source attribution and terms. The FX acquisition route is documented separately; source redistribution rights remain unresolved. Public new outputs contain aggregates and numerical diagnostics, while per-origin financial arrays remain private. A later download of the same series can differ from the frozen version.

The original ridge optimizer uses two-block ADMM (Boyd et al., 2011). Acceptance requires finite calculations, the declared primal/dual residual conditions and a feasible-dual objective-gap check. Its floating-point gap certificate bounds optimization error in the stated objective, rather than statistical adequacy or issued-report stability. An unpenalized intercept can have a flat minimizer interval: at real $\tau = 19/20$, nineteen zeros and a single positive value M admit every intercept in $[0, M]$. The implementation documentation retains the dual construction, iteration limits, tolerances and resource checks.

E1–E4 form a later exploratory supplement. The inference audit records the fixed Holm family, bandwidth sensitivity and coverage initialization correction. E3 records its pre-run menu and resource limits, original candidate eligibility, exact tie handling and all outcomes. The earlier four-predictor outputs were reused after reconciliation with their stored arrays and validation selections. These checks establish what was computed on the retained inputs; they add no independent financial sample.

The former long coverage-evidence table, full loss-weight frontiers, machine diagnostics and execution histories remain in their original repository paths. The result archive retains adverse cells and original file identities. New coverage diagnostics supplement those records. The new manuscript is compiled directly from the delivered TeX using the bundled Tectonic engine, and the supplied PDF is that compiler's output.